



UNIVERSITAT
POLITÈCNICA
DE VALÈNCIA



Session 0. Introduction

Aplicaciones de Reconocimiento de Formas (ARF)

Curso 2022/2023

Departamento de Sistemas Informáticos y Computación

Objectives

- To develop fully-fledge systems in which pattern recognition (PR) techniques, algorithms and tools previously studied are integrated.
- To provide a practical view to build PR systems.
- To present tools not covered in previous courses.
- More precisely, tools on the following fields will be introduced:
 - Text and image processing with CRF: CRFSuite
 - Neural networks: PyTorch and HuggingFace

Schedule

01/03/23: Intro + CRFSuite (Jorge + Carlos)

08/03/23: PyTorch (Carlos)

22/03/23: HuggingFace (Jorge)

29/03/23: HuggingFace (Jorge)

05/04/23: HuggingFace (Jorge)

19/04/23: Proposal and assignment of projects, working on the project

26/04/23: Working on the project

03/05/23: Project review (20% of evaluation)

10/05/23: Working on the project

17/05/23: Working on the project

31/05/23: Final presentation of the project - Day 1 (80% of evaluation)

01/06/23: Final presentation of the project - Day 2 (80% of evaluation)

Selected Students' Projects 2022

- Starch fast detection in augmented images of honey with yolov5
- Clasificación de papers de ML utilizando GNNs.
- Fast Ray Tracing IA Denosing
- Sentence Embedding para fine-tuning adaptativo
- Descripción automática de imágenes
- Predicting Grasp Stability with Tactile Sensors
- Predicción de frames futuros en secuencias animadas mediante CRF
- Detección de la utilidad en los comentarios de profesores y alumnos
- Segmentación 3D de gliomas con redes neuronales convolucionales
- Multimedia Automatic Misogyny Identification
- X-ray Chest Classification Task using ResNet, Inception and Transformed based
- Detección de Alzheimer mediante clasificación de imágenes de resonancias magnéticas
- Clasificador de gemas
- Estimación automática de biomasa de especies marinas con visión por computador
- Detección facial de mascarilla
- Clasificación de Variantes genéticas empleando redes neuronales
- Deteccion y clasificación de objetos para conduccion autónoma
- Reinforcement Learning aplicado a Lunar Lander
- Clasificación de toxicidad de medicamentos

Work space

- Boot your virtual machine (VM) from <https://portal-ng.dsic.upv.es>
- Software, scripts and data are available in your VM
`rdesktop -u arf ARF-$USER-STUDENT-2023.dsicv.upv.es`
add “-f” for fullscreen mode or “-g 90%” to use 90% of the whole screen
- Default password is arf2023, please change it immediately.
- You are sudoer in your VM, so you can install additional software
- ARF VMs are 1-core machines with 2 GB RAM and 50 GB HDD
- Intended for experiments on small datasets and low computational demand
- Further details to connect from home, check manual in PoliformaT